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Electronics Editorial Office
Young Investigator Award Committee
MDPI, St. Alban-Anlage 66, 4052 Basel, Switzerland

Dear Members of the Award Committee,

Cover letter in support of my nomination for the Electronics Young Investigator Award 2026

I am writing in support of my nomination for the *Electronics* Young Investigator Award. My research develops **physics-informed artificial intelligence and hardware-validated digital twins for power-electronic and electromechanical systems**, an area that sits within the journal's scope across its Artificial Intelligence, Power Electronics, and Industrial Electronics sections. I completed my PhD in Electrical Engineering at Southeast University in September 2025 and am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham.

The contribution I would highlight is a single, coherent capability: turning the slow, high-fidelity physics of electronic and electromechanical systems into **fast, accurate, and differentiable models that can be used in real time**. At the centre of this work is a **deep-learning surrogate** that maps design parameters to the full electromagnetic field of an electric machine, reaching $R^2 = 0.9996$ for average torque on unseen designs with **more than 94% inference speed-up** over finite-element analysis. Because it is implemented in PyTorch, its gradients are available end-to-end, which makes it directly usable for sensitivity analysis and gradient-based design. This is, in substance, the computational engine of a **real-time digital twin** of a power-electronic drive.

This AI layer rests on a foundation of **validated multiphysics modelling**. I developed a high-fidelity thermal model with fluid-solid coupled cooling that predicts temperature fields in seconds where an equivalent simulation needs many CPU-hours, validated against dynamometer measurements across the full operating map. I have also studied **PWM excitation and converter-driven loss** in electric drives, and I hold **9 patents**, several of which concern circuit and control methods for suppressing induced-voltage ripple and torque pulsation in power-electronic machine drives. Together these works connect circuit-level excitation, machine behaviour, and AI-based acceleration into one design-to-operation pipeline.

The research has been recognised and carried into practice. My first-author work received the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**, and in the same year my electric-drive design won **1st Prize in the IEEE TEC Electric Vehicle Design Competition**. Working with **Inovance**, a world-leading traction-machine supplier, I designed, built, and tested two high-speed oil-cooled drive systems (up to 270 kW, 97.4% efficiency at 20,000 rpm) through the full industrial cycle. My record comprises **23 journal papers** (19 in IEEE Transactions; 11 first-author), 18 conference papers, and 9 patents.

I am an early-career researcher, and I see this work as the beginning of a longer programme on real-time, physics-informed digital twins for electronic systems, applied to energy-efficient and reliable power electronics. I am grateful to my nominators for their support, and I would be glad to provide any further material the Committee may find useful. Thank you for considering my nomination.

Yours sincerely,

Wentao Zhang